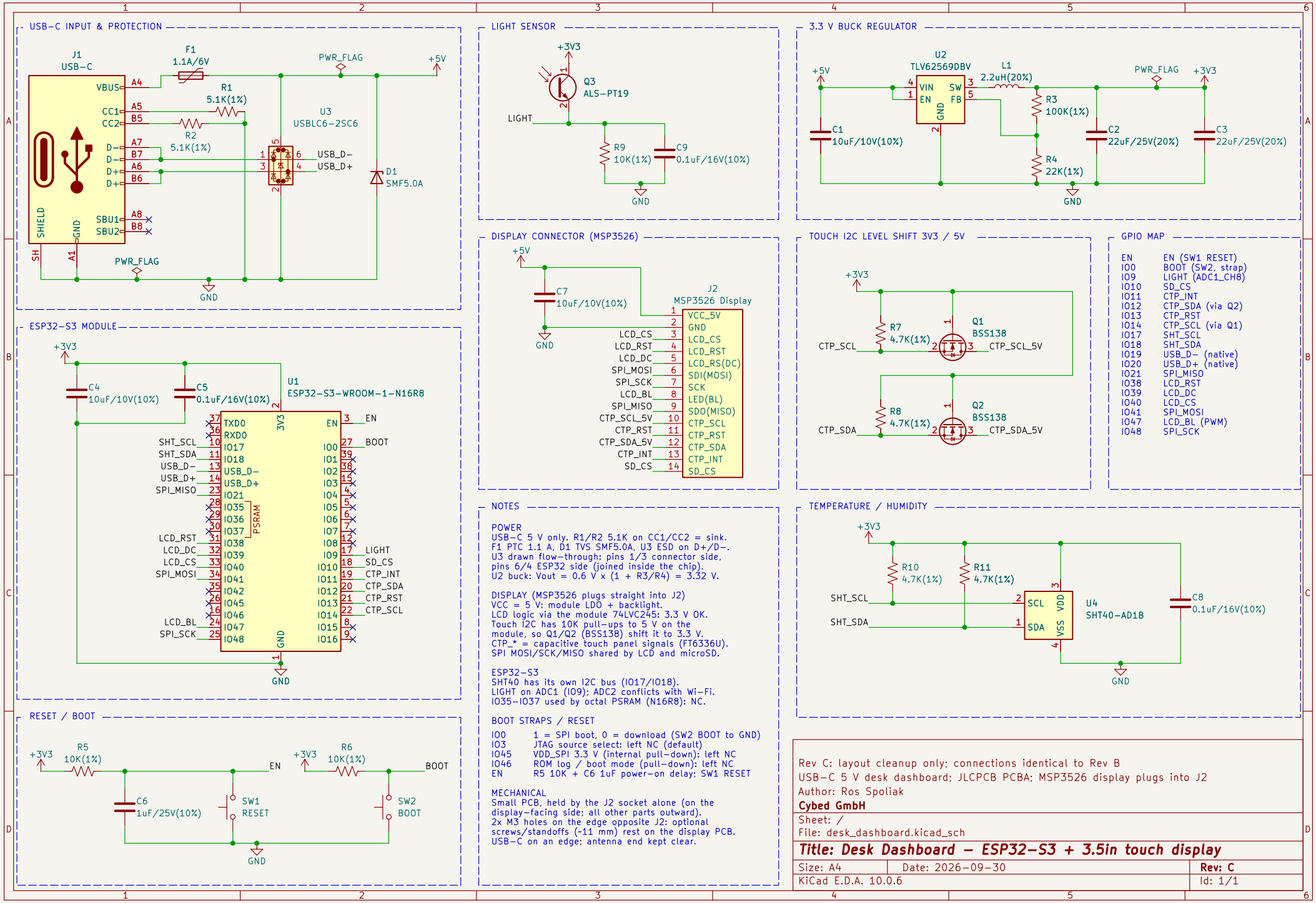

USB-C INPUT & PROTECTION
J1
USB-C
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
SH
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
F1
1.1A/6V
R1
5.1K(1%)
R2
5.1K(1%)
U3
USBLC6-2SC6
USB_D-
USB_D+
D1
SMF5.0A
PWR_FLAG
+5V
PWR_FLAG
GND
LIGHT SENSOR
+3V3
Q3
ALS-PT19
LIGHT
R9
10K(1%)
C9
0.1uF/16V(10%)
GND
3.3 V BUCK REGULATOR
U2
TLV62569DBV
+5V
VIN
EN
SW
FB
GND
L1
2.2uH(20%)
R3
100K(1%)
R4
22K(1%)
C1
10uF/10V(10%)
C2
22uF/25V(20%)
C3
22uF/25V(20%)
PWR_FLAG
+3V3
GND
DISPLAY CONNECTOR (MSP3526)
+5V
C7
10uF/10V(10%)
GND
J2
MSP3526 Display
VCC_5V
GND
LCD_CS
LCD_RST
LCD_DC
SPI_MOSI
SPI_SCK
LCD_BL
SPI_MISO
CTP_SCL_5V
CTP_RST
CTP_SDA_5V
CTP_INT
SD_CS
LCD_RS(DC)
SDI(MOSI)
SCK
LED(BL)
SDO(MISO)
CTP_SCL
CTP_RST
CTP_SDA
CTP_INT
SD_CS
TOUCH I2C LEVEL SHIFT 3V3 / 5V
+3V3
R7
4.7K(1%)
Q1
BSS138
CTP_SCL
CTP_SCL_5V
R8
4.7K(1%)
Q2
BSS138
CTP_SDA
CTP_SDA_5V
GPIO MAP
EN EN (SW1 RESET)
IO0 BOOT (SW2, strap)
IO9 LIGHT (ADC1_CH8)
IO10 SD_CS
IO11 CTP_INT
IO12 CTP_SDA (via Q2)
IO13 CTP_RST
IO14 CTP_SCL (via Q1)
IO17 SHT_SCL
IO18 SHT_SDA
IO19 USB_D- (native)
IO20 USB_D+ (native)
IO21 SPI_MISO
IO38 LCD_RST
IO39 LCD_DC
IO40 LCD_CS
IO41 SPI_MOSI
IO47 LCD_BL (PWM)
IO48 SPI_SCK
ESP32-S3 MODULE
+3V3
C4
10uF/10V(10%)
C5
0.1uF/16V(10%)
U1
ESP32-S3-WROOM-1-N16R8
TXD0
RXD0
3V3
EN
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
PSRAM
SHT_SCL
SHT_SDA
USB_D-
USB_D+
SPI_MISO
LCD_RST
LCD_DC
LCD_CS
SPI_MOSI
LCD_BL
SPI_SCK
EN
BOOT
LIGHT
SD_CS
CTP_INT
CTP_SDA
CTP_RST
CTP_SCL
GND
TEMPERATURE / HUMIDITY
+3V3
R10
4.7K(1%)
R11
4.7K(1%)
SHT_SCL
SHT_SDA
U4
SHT40-AD1B
SCL
SDA
VDD
VSS
C8
0.1uF/16V(10%)
GND
RESET / BOOT
+3V3
R5
10K(1%)
EN
C6
1uF/25V(10%)
SW1
RESET
+3V3
R6
10K(1%)
BOOT
SW2
BOOT
GND
NOTES
POWER
USB-C 5 V only. R1/R2 5.1K on CC1/CC2 = sink.
F1 PTC 1.1 A, D1 TVS SMF5.0A, U3 ESD on D+/D-.
U3 drawn flow-through: pins 1/3 connector side,
pins 6/4 ESP32 side (joined inside the chip).
U2 buck: Vout = 0.6 V x (1 + R3/R4) = 3.32 V.
DISPLAY (MSP3526 plugs straight into J2)
VCC = 5 V: module LDO + backlight.
LCD logic via the module 74LVC245: 3.3 V OK.
Touch I2C has 10K pull-ups to 5 V on the
module, so Q1/Q2 (BSS138) shift it to 3.3 V.
CTP_* = capacitive touch panel signals (FT6336U).
SPI MOSI/SCK/MISO shared by LCD and microSD.
ESP32-S3
SHT40 has its own I2C bus (IO17/IO18).
LIGHT on ADC1 (IO9); ADC2 conflicts with Wi-Fi.
IO35-IO37 used by octal PSRAM (N16R8): NC.
BOOT STRAPS / RESET
IO0 1 = SPI boot, 0 = download (SW2 BOOT to GND)
IO3 JTAG source select: left NC (default)
IO45 VDD_SPI 3.3 V (internal pull-down): left NC
IO46 ROM log / boot mode (pull-down): left NC
EN R5 10K + C6 1uF power-on delay; SW1 RESET
MECHANICAL
Small PCB, held by the J2 socket alone (on the
display-facing side; all other parts outward).
2x M3 holes on the edge opposite J2: optional
screws/standoffs (~11 mm) rest on the display PCB.
USB-C on an edge; antenna end kept clear.
Rev C: layout cleanup only; connections identical to Rev B
USB-C 5 V desk dashboard; JLCPCB PCBA; MSP3526 display plugs into J2
Author: Ros Spoliak
Cybed GmbH
Sheet: /
File: desk_dashboard.kicad_sch
Title: Desk Dashboard – ESP32-S3 + 3.5in touch display
Size: A4
Date: 2026-09-30
Rev: C
KiCad E.D.A. 10.0.6
Id: 1/1